

## WHAT THIS IS FOR

Checking that a PCR produced a product and that it is the right size. You are **not purifying** — nothing is recovered from this gel.

Slabs are **1% agarose in 1× TAE**, cut from the pre-made gels in the fridge.

## PREPARE EACH SAMPLE

**8** loading dye (tube marked *load*)

**3** PCR product

Mix, quick spin.

**Marker, one per section:** 8 µL loading dye into a tube of marker (yellow), mix, spin.

## SET UP AND LOAD

1. Cut a slab with enough wells for every sample, the marker, and a few spare.
2. Put it in the rig. Fill with **1× TAE** until it is just covered.
3. Brace the gel with the plastic wedge so it cannot float.
4. Write the sample order on a paper strip; line the tubes up to match.
5. Load **9 µL** per well with a P20.

## RUN

**Run to red.** DNA is negative and moves to the red electrode, so load at the black end.

Run at **175 V**. It takes about **10 min**.

**Do not set a timer and walk away.** Check the gel **every minute or so** and watch how the blue band is progressing. Stop it when the front is **2/3–3/4 down the gel** — that is the endpoint, not the clock.

## IMAGE

1. Move the gel to the imager and lay the label strip above it.
2. Photograph it **with your phone through the orange filter**.
3. Name the file by date and time, then upload it.

## READING IT

- Smaller fragments move faster.
- A **library** gives a smear, not a sharp band — its members differ in length.
- Loading dye does two jobs: glycerol sinks the sample into the well, and the colour tracks the run. It does not bind DNA.
- Load below the full volume on purpose — 9 µL of 11 µL — so you do not draw an air gap.
- Do not puncture a well. The sample runs out the bottom; use another lane.